Heterogeneous Beliefs and Bubbles

Harrison–Kreps (1978) — QuantEcon Lecture 89

Thomas J. Sargent John Stachurski

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Outline

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Overview

The Harrison–Kreps Model

Determines the **equilibrium price** of a dividend-yielding asset traded by two types of self-interested investors.

The model features:

- Heterogeneous beliefs about dividends
- Incomplete markets
- Short-sales constraints
- (Possibly) leverage limits on borrowing to finance asset purchases

Reference

Harrison, J. M. and Kreps, D. M. (1978). "Speculative Investor Behavior in a Stock Market with Heterogeneous Expectations." *Quarterly Journal of Economics*, 92(2), 323–336.

What Is a Bubble?

Economists differ on definitions. The Harrison–Kreps model illustrates:

A Working Definition of a Bubble

A component of an asset price can be interpreted as a **bubble** when *all* investors agree that the current price of the asset exceeds what they believe the asset's underlying dividend stream justifies.

We will see that this can arise in equilibrium when

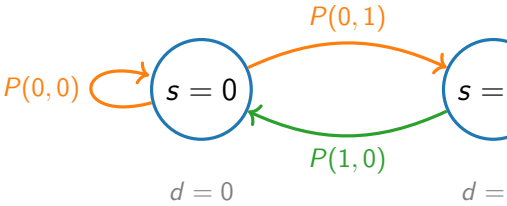
1. Investors have *heterogeneous* beliefs, and
2. Short sales are *prohibited* (pessimists cannot short).

Asset and Dividends

- Fixed supply of A shares of a risky asset.
- Dividends $\{d_t\}$ follow a **two-state Markov chain**:

$$S = \{0, 1\}, \qquad d_t = \begin{cases} 0 & s_t = 0 \\ 1 & s_t = 1 \end{cases}$$

- Stock is traded **ex dividend**: owning a share at end of t entitles the holder to the dividend at $t+1$ and the right to sell at $t+1$.



Two Types of Investors

Two types $h \in \{a, b\}$ of risk-neutral investors share a **common discount factor** $\beta = 0.75$ but hold **different beliefs** about the Markov transition matrix, $P(i, j) = \Pr\{s_{t+1} = j \mid s_t = i\}$:

Type a believes

$$P_a = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

Type b believes

$$P_b = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix}$$

- In state 0: type a is **more optimistic** about tomorrow's dividend than type b $\left[\frac{1}{2} > \frac{1}{3}\right]$
- In state 1: type b is **more optimistic** about tomorrow's dividend than type a $\left[\frac{3}{4} > \frac{1}{3}\right]$

Stationary Distributions

The **long-run** (stationary) distributions of the two transition matrices are:

Type *a*: stationary distribution

$$\pi_a \approx \begin{bmatrix} 0.571 & 0.429 \end{bmatrix}$$

Type *a* spends more time in the *low-dividend* state.

$\Rightarrow$ **more pessimistic on average**.

Type *b*: stationary distribution

$$\pi_b \approx \begin{bmatrix} 0.429 & 0.571 \end{bmatrix}$$

Type *b* spends more time in the *high-dividend* state.

$\Rightarrow$ **more optimistic on average**.

Key insight

Beliefs are *stochastically alternating*: each type is temporarily optimistic in one state and temporarily pessimistic in the other.

Ownership Rights, Short Sales, and Leverage

Ownership Rights

- Owning a share at end of $t \Rightarrow$ dividend at $t+1$ and right to sell at $t+1$.
- Both types are **risk-neutral**; common discount factor $\beta = 0.75$.

No Short Sales

- Pessimists **can** express their views by selling their shares.
- Pessimists **cannot** manufacture more shares by borrowing from optimists and immediately selling (no short selling).

Two Regimes to Study

1. Each type has *enough wealth* to purchase the entire stock (Harrison–Kreps's case).
2. Neither type alone can purchase the entire stock—both must hold some shares.

Information

- Investors know the current state s_t when they decide to buy or sell.
- They also know a **price function**

$$p : S \to \mathbb{R}_+, \quad s \mapsto p(s),$$

  which maps today's state into the equilibrium price.
- The price function is *endogenous*—it must be determined in equilibrium.

Summary of Equilibrium Prices

Scenario	**State 0**	**State 1**
p_a (homogeneous beliefs P_a)	1.33	1.22
p_b (homogeneous beliefs P_b)	1.45	1.91
p_o (optimistic marginal investor)	**1.85**	**2.08**
p_p (pessimistic marginal investor)	1.00	1.00
$\hat{p}_a$ (willingness to pay, type a)	1.85	1.69
$\hat{p}_b$ (willingness to pay, type b)	1.69	2.08

Parameters: $\beta = 0.75$; P_a, P_b as specified.

Single Belief (Homogeneous) Pricing

When all investors are of type h, the no-arbitrage pricing equation is:

$$p_h(s) = \beta\big[P_h(s,0)\,(0 + p_h(0)) + P_h(s,1)\,(1 + p_h(1))\big], \quad s = 0, 1.$$

In vector form:

$$\begin{bmatrix} p_h(0) \\ p_h(1) \end{bmatrix} = \beta\,[I - \beta P_h]^{-1} P_h \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Under P_a

$$p_a(0) = 1.33, \quad p_a(1) = 1.22$$

Under P_b

$$p_b(0) = 1.45, \quad p_b(1) = 1.91$$

These are the **fundamental values** as each type perceives them—the expected discounted present value of future dividends under that type's beliefs.

Pricing with Heterogeneous Beliefs: Optimistic Marginal Investor

Case: Both types have enough wealth to buy the entire stock; no short sales.

The marginal investor who *prices* the asset is the **temporarily optimistic** type. The equilibrium price $\bar{p}$ satisfies:

Harrison–Kreps Key Equation

$$\bar{p}(s) = \beta \max\{P_a(s,\cdot)\cdot[\bar{p}+d],\ P_b(s,\cdot)\cdot[\bar{p}+d]\}, \quad s = 0, 1,$$

where $d = \left[\begin{smallmatrix}0\\1\end{smallmatrix}\right]$ and $P_h(s,\cdot)\cdot[\bar{p}+d] = P_h(s,0)\,\bar{p}(0) + P_h(s,1)\,(1+\bar{p}(1))$.

- The max selects whichever type values the asset more highly *next period*.
- Type a prices in state 0; type b prices in state 1.

Solving the Functional Equation: Value Iteration

Like a Bellman equation, iterate to convergence:

$$\bar{p}^{j+1}(s) = \beta \max\left\{ P_a(s,0)\,\bar{p}^j(0) + P_a(s,1)\,(1+\bar{p}^j(1)),\ P_b(s,0)\,\bar{p}^j(0) + P_b(s,1)\,(1+\bar{p}^j(1)) \right\}$$

starting from any initial guess, e.g. $\bar{p}^0 = (0,0)$.

This operator is a **contraction mapping** $\Rightarrow$ unique fixed point.

Optimistic equilibrium prices

$$p_o(0) = 1.85, \qquad p_o(1) = 2.08.$$

Both exceed $p_a(s)$ and $p_b(s)$ in every state!

Why Does the Bubble Arise?

Each investor's **willingness to pay** (holding value):

Type a

$$\hat{p}_a(s) = \begin{cases} \bar{p}(0) & s = 0 \\ \beta\big[P_a(1,0)\,\bar{p}(0) \\ \quad + P_a(1,1)(1+\bar{p}(1))\big] & s = 1 \end{cases}$$

$\hat{p}_a(0) = 1.85, \quad \hat{p}_a(1) = 1.69$

Type b

$$\hat{p}_b(s) = \begin{cases} \beta\big[P_b(0,0)\,\bar{p}(0) \\ \quad + P_b(0,1)(1+\bar{p}(1))\big] & s = 0 \\ \bar{p}(1) & s = 1 \end{cases}$$

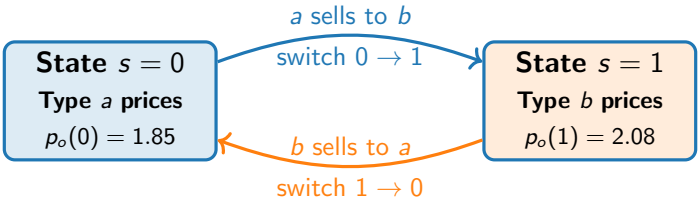
$\hat{p}_b(0) = 1.69, \quad \hat{p}_b(1) = 2.08$

Note: $\hat{p}_a(1) = 1.69 < \bar{p}(1) = 2.08$ and $\hat{p}_b(0) = 1.69 < \bar{p}(0) = 1.85$.

The resale-option premium

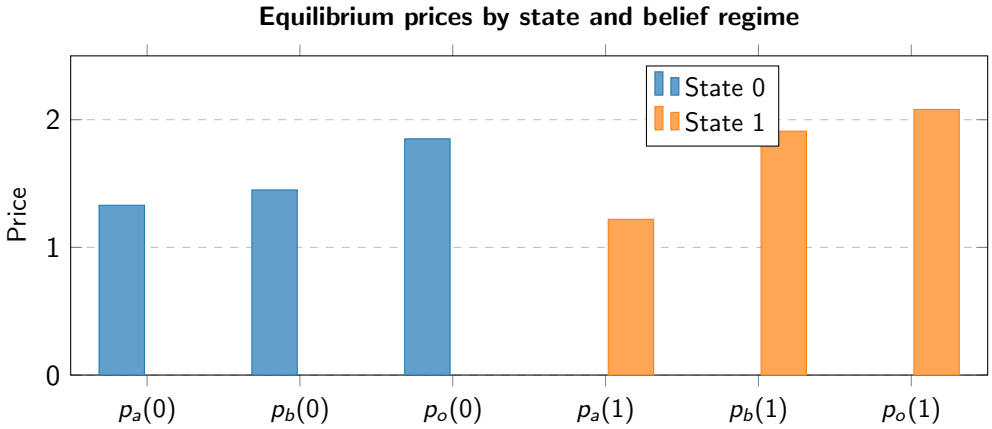
Each investor pays *more* than their own fundamental value because they expect to resell to a more optimistic buyer later.

Asset Trading Dynamics



- The asset **changes hands** every time the state switches.
- High trading volume is a signature feature of this bubble model.
- Even the *pessimistic* type at each state acknowledges the price exceeds its own fundamental value—yet still holds the asset.

Visualizing the Price Premium



The optimistic equilibrium price p_o exceeds both homogeneous belief benchmarks p_a and p_b in every state.

Pricing with Pessimistic Marginal Investor: Insufficient Funds

Case: The optimistic type *cannot* purchase the entire stock; both types must hold shares. The asset price must attract pessimistic investors too.

Pessimistic Pricing Equation

$$\check{p}(s) = \beta \min\{P_a(s,\cdot)\cdot[\check{p}+d],\ P_b(s,\cdot)\cdot[\check{p}+d]\}, \quad s = 0, 1.$$

Pessimistic equilibrium prices

$$p_p(0) = 1.00, \qquad p_p(1) = 1.00.$$

- $p_p < p_a, p_b$ in every state—below both fundamental values.
- Optimistic investors think the asset is *underpriced* but are prevented from leveraging up to buy more.
- Pessimistic pricing also arises if short sales *are* allowed: pessimists short the asset until price falls to $\check{p}$.

Scheinkman's Interpretation

Jose Scheinkman (2014) interprets Harrison–Kreps as a model of a bubble: an asset price that exceeds *every* investor's estimate of fundamental value.

Key predictions of the model:

- **High volume** accompanies high (optimistic) prices—the asset changes hands with every state switch.
- Bubbles are supported by short-sales constraints and leverage.
- Bubbles **end** when asset supply grows enough to exceed optimistic investors' resources.

Regulatory implications

- **Limiting short sales** $\uparrow$ bubble prices (pessimists can't short).
- **Limiting leverage** $\downarrow$ bubble prices (optimists can't borrow to buy).
- These two regulations have *opposite* effects!

Permanently Optimistic and Pessimistic Investors

To interpret p_o further, introduce two hypothetical special types:

Permanently optimistic investor

Holds the most optimistic view in *each* state:

$$P_{\text{opt}} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} \end{bmatrix}$$

(Takes the max column from P_a and P_b row by row.)

Permanently pessimistic investor

Holds the least optimistic view in *each* state:

$$P_{\text{pess}} = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

(Takes the min column from P_a and P_b row by row.)

Key result

The equilibrium price under heterogeneous beliefs with optimistic marginal investor, p_o, **equals** the single-belief price of the *permanently optimistic* investor.

Extended Summary Table

Scenario	State 0	State 1
p_a (type a only)	1.33	1.22
p_b (type b only)	1.45	1.91
p_{opt} (perm. opt.)	1.85	2.08
p_{pess} (perm. pess.)	1.33	1.00
p_o (optimistic marginal inv.)	**1.85**	**2.08**
p_p (pessimistic marginal inv.)	1.00	1.00
$\hat{p}_a$ (type a WTP)	1.85	1.69
$\hat{p}_b$ (type b WTP)	1.69	2.08

$p_o = p_{\mathrm{opt}}$: the heterogeneous-beliefs equilibrium replicates the price of a permanently-optimistic single-belief market.

Summary

1. **Heterogeneous beliefs + no short sales = bubble.** Prices can exceed every investor's fundamental value.
2. **The resale-option premium.** Investors overpay because they anticipate selling to a more optimistic buyer when the state changes.
3. **Trading volume is high** along the bubble path—the asset changes hands every time the Markov state switches.
4. **The optimistic equilibrium price equals the single-belief price of a permanently-optimistic investor.**
5. **Regulatory asymmetry:** limiting short sales *inflates* bubbles; limiting leverage *deflates* them.
6. **Supply matters:** expanding asset supply beyond optimists' financial capacity extinguishes the bubble.

References

- Harrison, J. M. and Kreps, D. M. (1978). "Speculative Investor Behavior in a Stock Market with Heterogeneous Expectations." *Quarterly Journal of Economics*, **92**(2), 323–336.
- Scheinkman, J. A. (2014). *Speculation, Trading, and Bubbles*. Columbia University Press.
- Sargent, T. J. and Stachurski, J. (2025). "Heterogeneous Beliefs and Bubbles." *QuantEcon Lecture 89*. `https://python.quantecon.org/harrison_kreps.html`